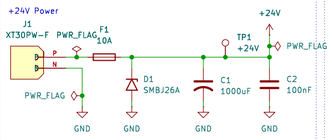
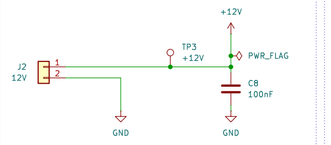
Power Circuits

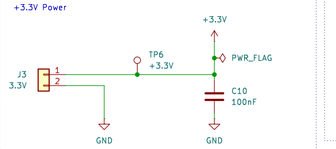
+24V Power



+12V Power

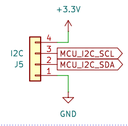


+3.3V Power

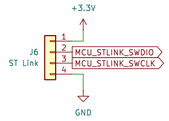


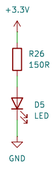


External Interface I2C



External Interface ST Link

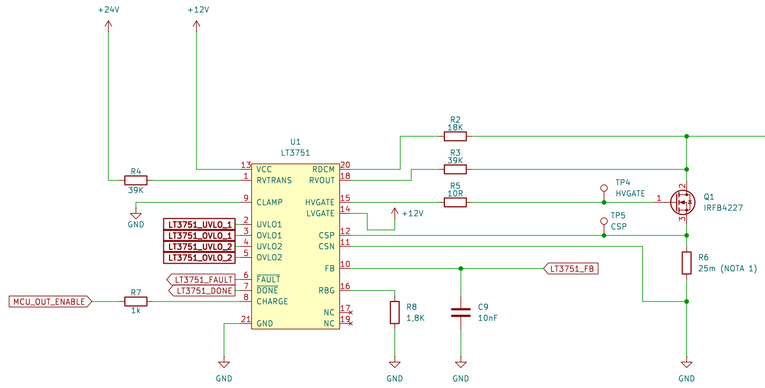


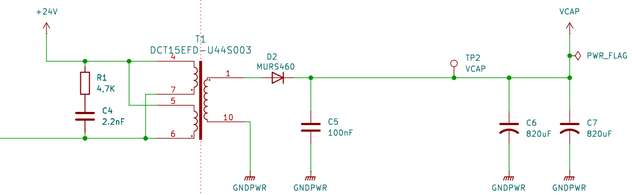


Mounting Holes



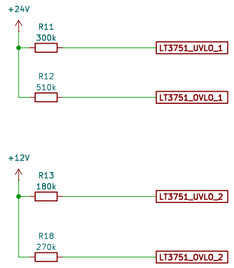
Main Circuit



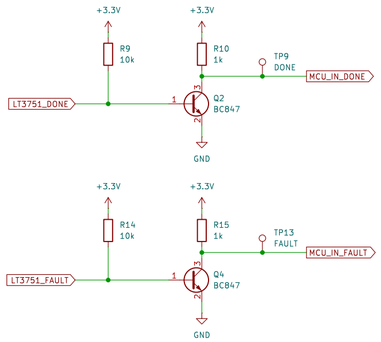


High Voltage

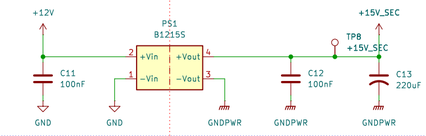
Undervoltage and Overvoltage LT3751 Detection



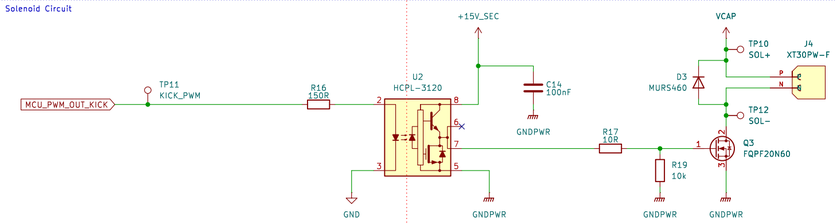
DONE and FAULT Signal Conditioning



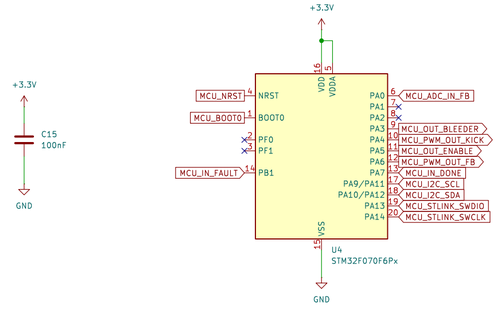
Isolated Power +15V



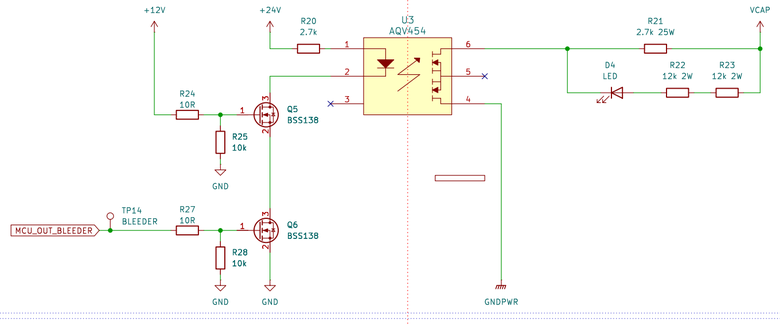
Solenoid Circuit



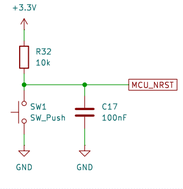
Microcontroller



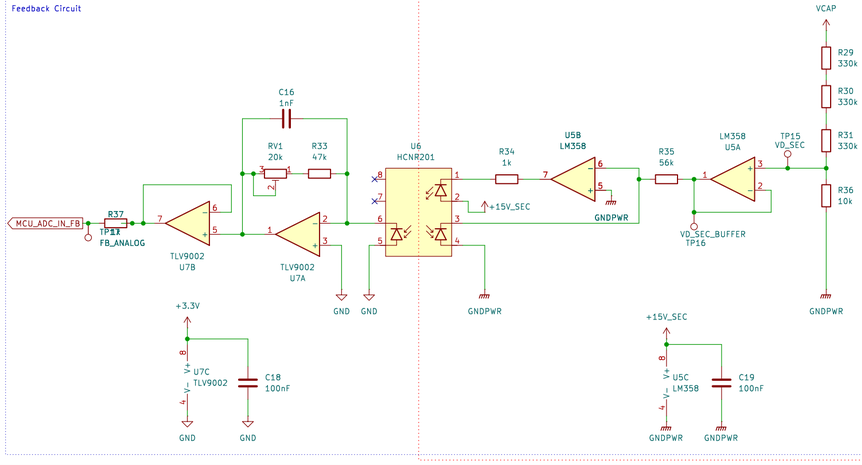
Automatic Bleeder



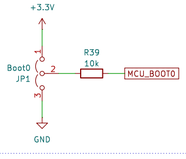
Reset Switch



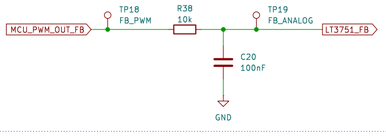
Feedback Circuit



BOOT0



Feedback PWM to Analog Low-Pass Filter



Notas:

1. +24V é fornecido por um pack Li-Ion 6S. A alimentação pode variar entre 16.8 e 25.2V.
2. R18 deve ter encapsulamento 1205. A resistência deve ser escolhida de acordo com a corrente máxima no primário do transformador, conforme relação a seguir. A potência do resistor deve ser acima da potência máxima dissipada, descrita abaixo:

* 50 mOhms (R025):	2,1 A	0,2 W
* 25 mOhms (R025):	4,2 A	0,4 W
* 20 mOhms (R020):	5,3 A	0,6 W
* 15 mOhms (R015):	7,1 A	0,7 W
* 12 mOhms (R012):	8,8 A	0,9 W
* 10 mOhms (R010):	10,6 A	1,1 W
* 6 mOhms (R006):	17,7 A	1,9 W

 Considerando I_pri(max) = 5 A para o transformador DCT15EFD-U44S003, adotar valores entre 20 e 50 mOhms.

3. Considerando V_out(trip) = 211V. Para outros valores de V_out(trip), consultar tabela.

Autor: Wildson Oliveira dos Santos
OxeBots

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Title: Projeto do Conversor Flyback para o Kicker

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